

ELITE MATHEMATICS

Pearson Edexcel International GCSE

Modular Unit 2

Adaptive Classified

OL-T51 · Vectors

With Answers

Questions with full worked solutions

■ 44 pages

FAMILY MAP

Vectors

F01	Use vectors with ratios and intersections	14 Q 67 marks
------------	--	-----------------

F02	Calculate with vectors	2 Q 5 marks
------------	-------------------------------	---------------

F03	Read and represent vectors	1 Q 1 marks
------------	-----------------------------------	---------------

23

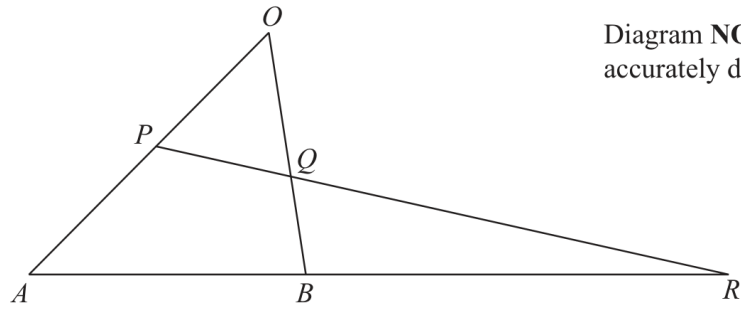


Diagram **NOT**
accurately drawn

OAB is a triangle.

P is the midpoint of OA

Q is a point on OB

ABR and PQR are straight lines.

$$\vec{OA} = 12\mathbf{a} \quad \vec{OB} = 8\mathbf{b}$$

(a) Express $\vec{AB}$ in terms of $\mathbf{a}$ and $\mathbf{b}$

(1)

Continued
4MA1-1H Q23

ORIGINAL QUESTION

23

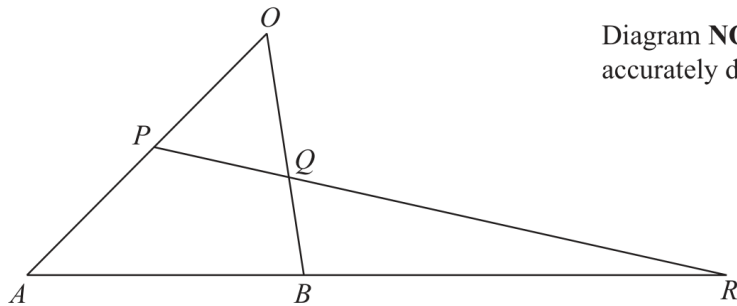


Diagram **NOT**
accurately drawn

OAB is a triangle.

P is the midpoint of OA

Q is a point on OB

ABR and PQR are straight lines.

$$\vec{OA} = 12\mathbf{a} \quad \vec{OB} = 8\mathbf{b}$$

$$AB:BR = 1:2 \quad \vec{OQ} = n\mathbf{b}$$

(b) Use a vector method to find the value of n

$$n = \dots\dots\dots (4)$$

Worked Solution 14

Vector method on two straight lines

UNIT 2
5 marks**Find AB**

$$\vec{AB} = \vec{OB} - \vec{OA} = 8\mathbf{b} - 12\mathbf{a}.$$

Find PR

$$\vec{OR} = \vec{OA} + 3\vec{AB} = -24\mathbf{a} + 24\mathbf{b}, \quad \vec{OP} = 6\mathbf{a}, \quad \vec{PR} = -30\mathbf{a} + 24\mathbf{b}.$$

Use collinearity

$$\vec{OQ} = \vec{OP} + \lambda\vec{PR} = 6\mathbf{a} + \lambda(-30\mathbf{a} + 24\mathbf{b}). \quad 6 - 30\lambda = 0 \Rightarrow \lambda = \frac{1}{5}, \quad n = 24\lambda = \frac{24}{5}.$$

Final answer

$$n = \frac{24}{5}$$

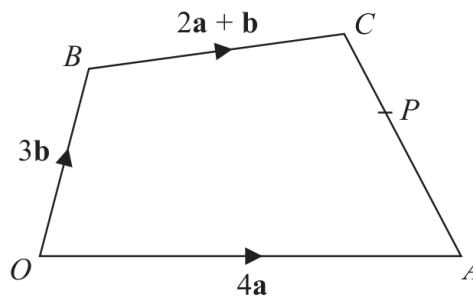
Question 26

4MA1-2H Q24

ORIGINAL QUESTION

6 marks

24

Diagram **NOT**
accurately drawnThe diagram shows a quadrilateral $OACB$ in which

$$\vec{OA} = 4\mathbf{a} \quad \vec{OB} = 3\mathbf{b} \quad \vec{BC} = 2\mathbf{a} + \mathbf{b}$$

- (a) Find $\vec{AC}$ in terms of $\mathbf{a}$ and $\mathbf{b}$
Give your answer in its simplest form.

$$\vec{AC} = \dots\dots\dots (2)$$

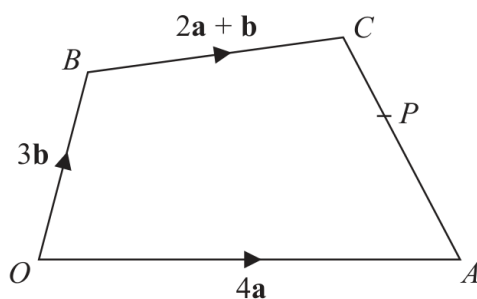
Question 26 (continued)

4MA1-2H Q24

ORIGINAL QUESTION

6 marks

24

Diagram **NOT**
accurately drawn

The diagram shows a quadrilateral $OACB$ in which

$$\vec{OA} = 4\mathbf{a} \quad \vec{OB} = 3\mathbf{b} \quad \vec{BC} = 2\mathbf{a} + \mathbf{b}$$

The point P lies on AC such that $AP:PC = 3:2$

The point Q is such that OPQ and BCQ are straight lines.

- (b) Using a vector method, find $\vec{OQ}$ in terms of $\mathbf{a}$ and $\mathbf{b}$
Give your answer in its simplest form.
Show your working clearly.

$$\vec{OQ} = \dots\dots\dots$$

(4)

Worked Solution 26

Vectors and collinearity

MODULAR UNIT 2

6 marks

Part (a)

$$\vec{AC} = 4b - 2a.$$

Point P

$$\vec{OP} = 4a + \frac{3}{5}(4b - 2a) = \frac{14}{5}a + \frac{12}{5}b.$$

Point Q

$$\vec{OQ} = \lambda \vec{OP} = 3b + \mu(2a + b) \Rightarrow \lambda = 3.$$

Final answer

$$\vec{OQ} = \frac{42}{5}a + \frac{36}{5}b$$

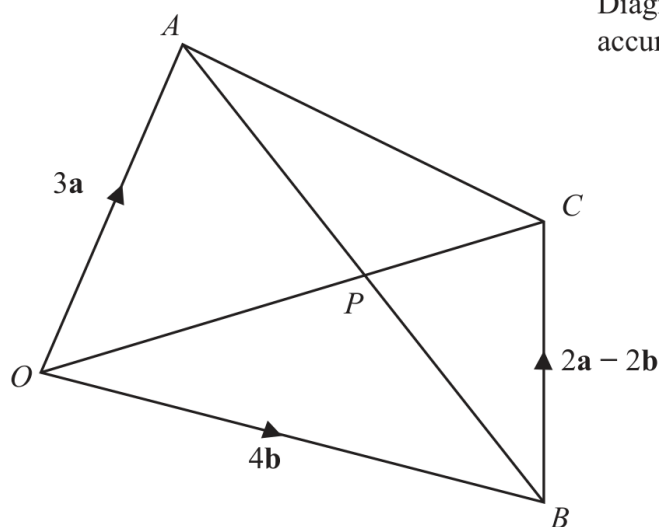
Question 35

4MA1-2H Q22 (a)(i), (a)(ii), (b)

ORIGINAL QUESTION UNIT 2

5 marks

22

Diagram NOT
accurately drawn $OACB$ is a quadrilateral.

$$\vec{OA} = 3\mathbf{a} \quad \vec{OB} = 4\mathbf{b} \quad \vec{BC} = 2\mathbf{a} - 2\mathbf{b}$$

- (a) (i) Find the vector $\vec{OC}$ in terms of $\mathbf{a}$ and $\mathbf{b}$
Simplify your answer.

$$\vec{OC} = \dots\dots\dots$$

(1)

- (ii) Find the vector $\vec{AB}$ in terms of $\mathbf{a}$ and $\mathbf{b}$

$$\vec{AB} = \dots\dots\dots$$

(1)

The point P lies on AB and on OC

- (b) Using a vector method, find the ratio $AP : PB$
Show your working clearly.

$$\dots\dots\dots$$

(3)

Worked solution 35

MODULAR UNIT 2

Find OC from two vectors / Find AB from position vectors / Ratio where two vector descriptions meet **5 marks****(a)(i) – Find OC from two vectors**

Use the path O-B-C

$$\overrightarrow{OC} = \overrightarrow{OB} + \overrightarrow{BC} = 4b + (2a - 2b) = 2a + 2b.$$

Final answer

$$OC = 2a + 2b$$

(a)(ii) – Find AB from position vectors

Subtract position vectors

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = 4b - 3a.$$

Final answer

$$AB = 4b - 3a$$

(b) – Ratio where two vector descriptions meet

Describe P on both lines

$$\overrightarrow{OP} = \lambda(2a + 2b) = 3a + \mu(4b - 3a).$$

Compare coefficients

$$2\lambda = 4\mu, \quad 2\lambda = 3 - 3\mu \Rightarrow \mu = \frac{3}{7}.$$

Convert the parameter to a ratio

$$AP : PB = \mu : (1 - \mu) = \frac{3}{7} : \frac{4}{7} = 3 : 4.$$

Final answer

$$AP:PB = 3:4$$

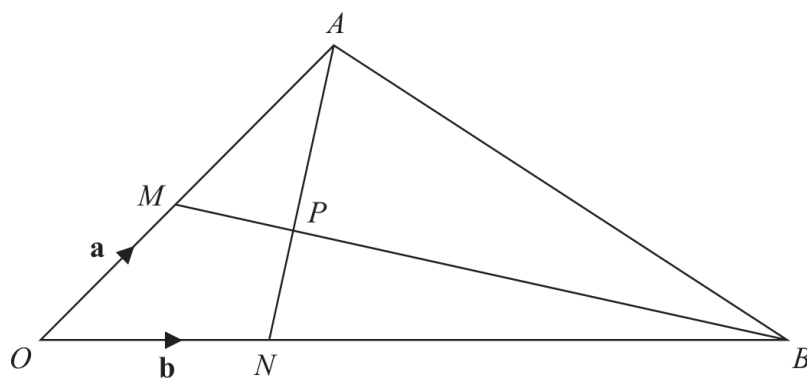
Question 15

4MA1-1H Q19

ORIGINAL QUESTION

5 marks

19

Diagram **NOT**
accurately drawn

OMA , ONB , MPB and NPA are straight lines.

M is the midpoint of OA

$ON:NB = 1:5$

$$\vec{OM} = \mathbf{a} \quad \vec{ON} = \mathbf{b}$$

(a) Find in terms of $\mathbf{a}$ and $\mathbf{b}$ the vector $\vec{AN}$

(1)

(b) Use a vector method to find the ratio $AP:PN$

$$AP:PN = \dots\dots\dots$$

(4)

Worked Solution 15

Intersection ratio by vectors

MODULAR UNIT 2

5 marks

Write the key vectors

$$\overrightarrow{OA} = 2\mathbf{a}, \quad \overrightarrow{OB} = 6\mathbf{b}, \quad \overrightarrow{AN} = \mathbf{b} - 2\mathbf{a}.$$

Describe P along AN

$$\overrightarrow{OP} = 2\mathbf{a} + \lambda(\mathbf{b} - 2\mathbf{a}) = (2 - 2\lambda)\mathbf{a} + \lambda\mathbf{b}.$$

Describe P along MB

$$\overrightarrow{OP} = \mathbf{a} + \mu(6\mathbf{b} - \mathbf{a}) = (1 - \mu)\mathbf{a} + 6\mu\mathbf{b}.$$

Equate coefficients

$$\lambda = 6\mu, \quad 2 - 2\lambda = 1 - \mu \Rightarrow \lambda = \frac{6}{11}.$$

Final answer

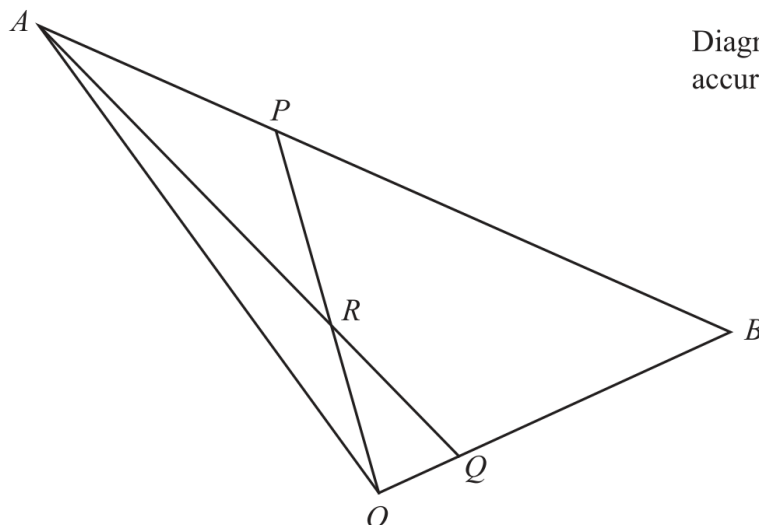
$$\overrightarrow{AN} = \mathbf{b} - 2\mathbf{a}; \quad AP : PN = 6 : 5$$

Original question 16

4MA1-1H Q24 M01, M02, M03

MODULAR UNIT 2

6 marks

24 OAB is a triangle.Diagram **NOT**
accurately drawn

$$\vec{OA} = 10\mathbf{a} \quad \vec{OB} = 10\mathbf{b}$$

 ARQ and ORP are straight lines.

$$\vec{AP} = \frac{1}{4} \vec{AB} \quad \text{and} \quad \vec{OQ} = \frac{1}{5} \vec{OB}$$

Write the following vectors in terms of $\mathbf{a}$ and $\mathbf{b}$
Simplify your answers.

(i) $\vec{AQ}$

.....
(1)

(ii) $\vec{OP}$

.....
(1)

Original question 16 (continued)

4MA1-1H Q24 M01, M02, M03

MODULAR UNIT 2

6 marks

(iii) $\vec{OR}$

(4)

Worked solution 16

Find AQ / Find OP / Find OR from two straight-line routes

MODULAR UNIT 2

6 marks

M01 – Find AQ**Use a route through O**

$$\overrightarrow{AQ} = \overrightarrow{AO} + \overrightarrow{OQ} = -10\mathbf{a} + 2\mathbf{b}.$$

Final answer

$$-10\mathbf{a} + 2\mathbf{b}$$

M02 – Find OP**Use AP=AB/4**

$$\overrightarrow{AP} = \frac{1}{4}(10\mathbf{b} - 10\mathbf{a}) = \frac{5}{2}\mathbf{b} - \frac{5}{2}\mathbf{a}.$$

Add OA

$$\overrightarrow{OP} = 10\mathbf{a} + \overrightarrow{AP} = \frac{15}{2}\mathbf{a} + \frac{5}{2}\mathbf{b}.$$

Final answer

$$\frac{15}{2}\mathbf{a} + \frac{5}{2}\mathbf{b}$$

M03 – Find OR from two straight-line routes**Write OR through P**

$$\overrightarrow{OR} = \overrightarrow{OP} + \lambda\overrightarrow{PR}.$$

Write OR through Q

Use the line ARQ and the known AQ route to form a second expression.

Equate coefficientsThe two routes give $\lambda=1/2$ and hence vector $\overrightarrow{OR}=(15)/(4)\mathbf{a}+5/4\mathbf{b}$.**Simplify**

$$\overrightarrow{OR} = \frac{15}{4}\mathbf{a} + \frac{5}{4}\mathbf{b}.$$

Final answer

$$\frac{15}{4}\mathbf{a} + \frac{5}{4}\mathbf{b}$$

Original question 29

4MA1-2H Q17 M01

MODULAR UNIT 2

3 marks

17 Here are two vectors.

$$\vec{FG} = \begin{pmatrix} -5 \\ 2 \end{pmatrix} \quad \vec{HG} = \begin{pmatrix} 4 \\ 14 \end{pmatrix}$$

Calculate the magnitude of the vector $\vec{HF}$

.....

Worked solution 29

Find the magnitude of HF

MODULAR UNIT 2

3 marks

M01 – Find the magnitude of HF**Find the vector**

$$\overrightarrow{HF} = \overrightarrow{HG} - \overrightarrow{FG} = (4, 14) - (-5, 2) = (9, 12).$$

Use Pythagoras

$$|HF| = \sqrt{9^2 + 12^2} = 15.$$

State the magnitude

$$|HF| = 15.$$

Final answer

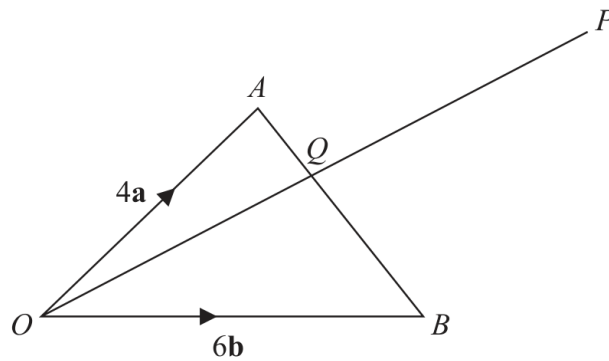
15

Original question 30

4MA1-2H Q21 Q21

ORIGINAL QUESTION**5 marks**

21

Diagram **NOT**
accurately drawn OAB is a triangle. Q is the point on AB such that OQP is a straight line.

$$\vec{OA} = 4\mathbf{a} \quad \vec{OB} = 6\mathbf{b} \quad \vec{AP} = 2\mathbf{a} + 8\mathbf{b}$$

Using a vector method, find the ratio $AQ:QB$

$$AQ:QB = \dots\dots\dots$$

Worked solution 30

Chapter: Vectors

MODULAR UNIT 2

5 marks

Q21 – Chapter: Vectors

Family: Use vectors with ratios and intersections • Variant: Recover a division ratio using vector parameters

Method

$$\mathbf{OP} = \mathbf{OA} + \mathbf{AP} = 6\mathbf{a} + 8\mathbf{b}.$$

Why: Add the consecutive displacement vectors.

Method

$$\text{Let } \mathbf{AQ} = t \mathbf{AB}. \text{ Then } \mathbf{OQ} = \mathbf{OA} + t(\mathbf{OB} - \mathbf{OA}) = (4 - 4t)\mathbf{a} + 6t\mathbf{b}.$$

Why: Parameterise Q along AB.

Method

$$\text{Since O, Q, P are collinear, } \mathbf{OQ} = \lambda(6\mathbf{a} + 8\mathbf{b}).$$

Why: Parallel vectors are scalar multiples.

Method

$$\text{Equating coefficients gives } 4 - 4t = 6\lambda \text{ and } 6t = 8\lambda, \text{ so } t = 8/17.$$

Why: Use independence of a and b to solve the two scalar equations.

Method

$$\text{Therefore } \mathbf{AQ}:\mathbf{QB} = (8/17):(9/17) = 8:9.$$

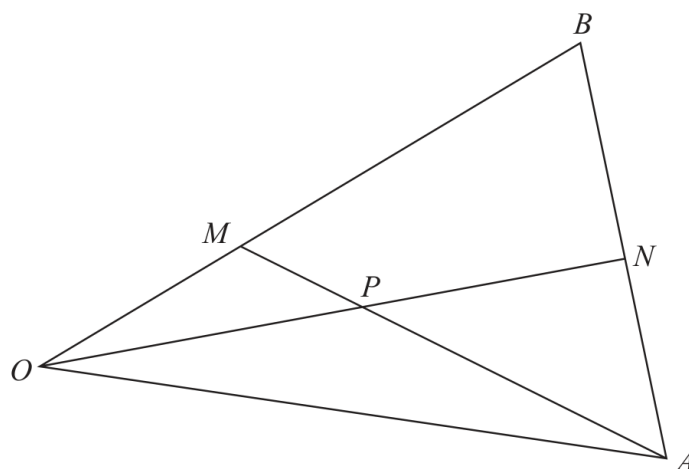
Why: The remaining fraction of AB is $1 - t$.

Final answer

$$\text{ratio: } \mathbf{AQ} : \mathbf{QB} = 8 : 9$$

Original question 13

4MA1-1H Q22

ORIGINAL QUESTION**5 marks****22** The diagram shows triangle OAB Diagram **NOT**
accurately drawn

$$\vec{OA} = 8\mathbf{a} \quad \vec{OB} = 6\mathbf{b}$$

 M is the point on OB such that $OM:MB = 1:2$ N is the midpoint of AB P is the point of intersection of ON and AM

Using a vector method, find $\vec{OP}$ as a simplified expression in terms of $\mathbf{a}$ and $\mathbf{b}$
 Show your working clearly.

$$\vec{OP} = \dots\dots\dots$$

Worked solution 13

Chapter: Vectors

MODULAR UNIT 2

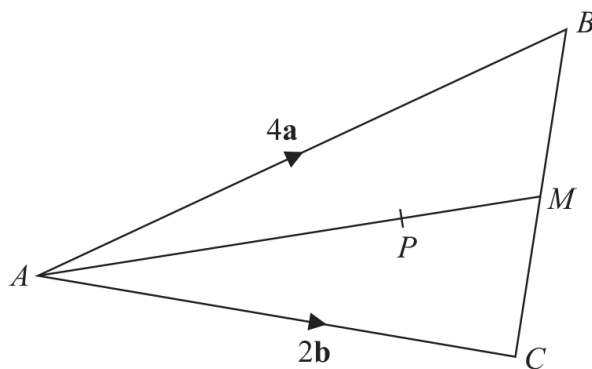
5 marks

Q22 – Chapter: Vectors

Family: Use vectors with ratios and intersections • Variant: Find an intersection vector from two paths

Method $OM = (1/3)OB = 2b$, so M has position vector $2b$.*Why: Use $OM:MB=1:2$.***Method**N is the midpoint of AB, so $ON = 4a + 3b$.*Why: Average $OA=8a$ and $OB=6b$.***Method**Let $OP = \lambda(4a + 3b)$ and also $OP = 8a + t(2b - 8a)$.*Why: P lies on both ON and AM.***Method**Comparing coefficients gives $8 - 8t = 4\lambda$ and $2t = 3\lambda$.*Why: Equate the a and b components.***Method**Solving gives $\lambda = 1/2$, so $OP = 2a + 3/2 b$.*Why: Substitute λ into $OP = \lambda ON$.***Final answer****OP:** $2a + (3/2)b$

23

Diagram **NOT**
accurately drawn ABC is a triangle.The midpoint of BC is M . P is a point on AM .

$$\vec{AB} = 4\mathbf{a}$$

$$\vec{AC} = 2\mathbf{b}$$

$$\vec{AP} = \frac{3}{2}\mathbf{a} + \frac{3}{4}\mathbf{b}$$

Find the ratio $AP:PM$

Worked solution

January 2020 | Question 23 | 3 marks

Family: Use vectors with ratios and intersections • Variant: Recover a division ratio using vector parameters

Method / working

1. $AB=4a$ and $AC=2b$, so $AM=2a+b$.
 2. $AP=(\frac{3}{2})a+(\frac{3}{4})b=(\frac{3}{4})(2a+b)=(\frac{3}{4})AM$.
 3. Therefore $PM=(\frac{1}{4})AM$ and $AP:PM=3:1$.
- Final answer: $AP:PM=3:1$

23 OAB is a triangle.

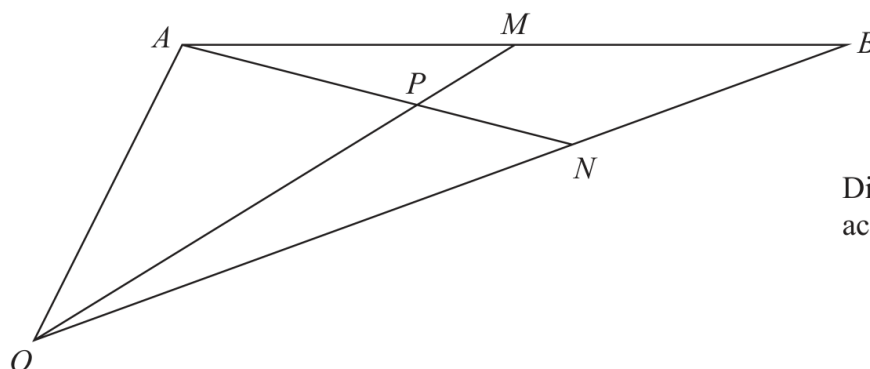


Diagram **NOT**
accurately drawn

$$\vec{OA} = 2\mathbf{a} \text{ and } \vec{OB} = 2\mathbf{b}$$

M is the midpoint of AB .

N is the point on OB such that $ON:NB = 2:1$

P is the point on AN such that OPM is a straight line.

Use a vector method to find $OP:PM$

Show your working clearly.

Worked solution

January 2021 | Question 23 | 6 marks

Family: Use vectors with ratios and intersections • Variant: Recover a division ratio using vector parameters

Method / working

1. Write vectors for M and N from $OA=2a$ and $OB=2b$.
2. Express OP along AN and also along OM because O,P,M are collinear.
3. Equate the a and b coefficients.
4. Solve the resulting scalar parameter.
5. Obtain $OP=(4/5)OM$.
6. Hence $PM=(1/5)OM$ and $OP:PM=4:1$.

Final answer: $OP:PM=4:1$

26 $OACB$ is a trapezium.

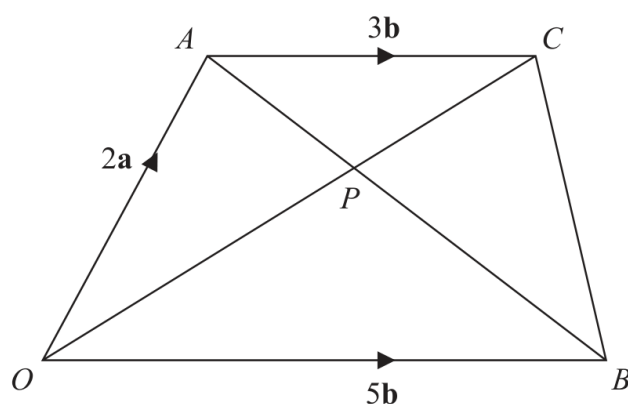


Diagram **NOT**
accurately drawn

$$\vec{OA} = 2\mathbf{a} \quad \vec{OB} = 5\mathbf{b} \quad \vec{AC} = 3\mathbf{b}$$

The diagonals, OC and AB , of the trapezium intersect at the point P .

Find and simplify an expression, in terms of $\mathbf{a}$ and $\mathbf{b}$, for $\vec{OP}$
Show your working clearly.

$$\vec{OP} = \dots\dots\dots$$

Worked solution

June 2021 | Question 26 | 5 marks

Family: Use vectors with ratios and intersections • Variant: Find an intersection vector from two paths

Method / working

1. Because P lies on OC, write $OP = n(2a + 3b)$.
2. Because P lies on AB, write a second vector equation using $OA = 2a$ and $OB = 5b$.
3. Equate coefficients of a and b to solve the scalar $n = 5/8$.
4. Substitute n into $OP = (5/8)(2a + 3b)$.
5. Simplify: $OP = (5/4)a + (15/8)b$.

Final answer: $(5/4)a + (15/8)b$

14 $ABCDEF$ and $GHIJKL$ are regular hexagons each with centre O .

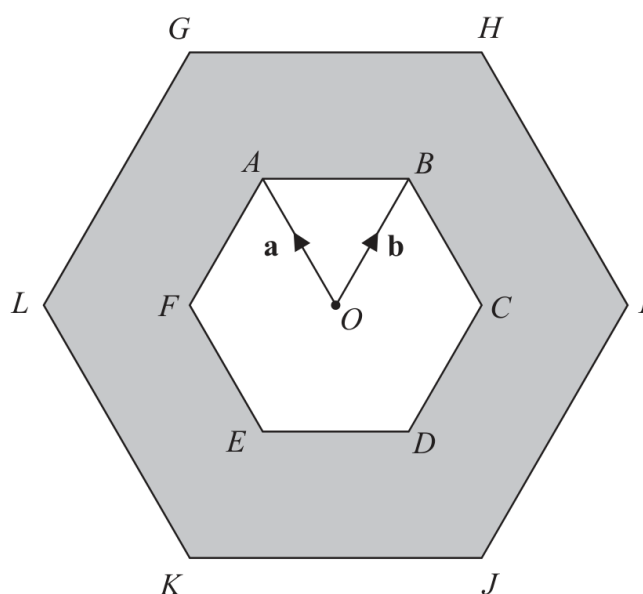


Diagram **NOT**
accurately drawn

$GHIJKL$ is an enlargement of $ABCDEF$, with centre O and scale factor 2

$$\vec{OA} = \mathbf{a} \quad \vec{OB} = \mathbf{b}$$

(a) Write the following vectors, in terms of $\mathbf{a}$ and $\mathbf{b}$.
Simplify your answers.

(i) $\vec{AB}$

(1)

(ii) $\vec{KI}$

(2)

(iii) $\vec{LD}$

(2)

The triangle OAB has an area of 5 cm^2

(b) Calculate the area of the shaded region.

(3)

cm^2

Worked solution

November 2021 | Question 14 | 1 marks

Family: Read and represent vectors • Variant: Express a directed segment in basis vectors

Method / working

1. $\overrightarrow{OA} = \mathbf{a}$, $\overrightarrow{OB} = \mathbf{b}$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

Final answer: (i) $\mathbf{b} - \mathbf{a}$

14 $ABCDEF$ and $GHIJKL$ are regular hexagons each with centre O .

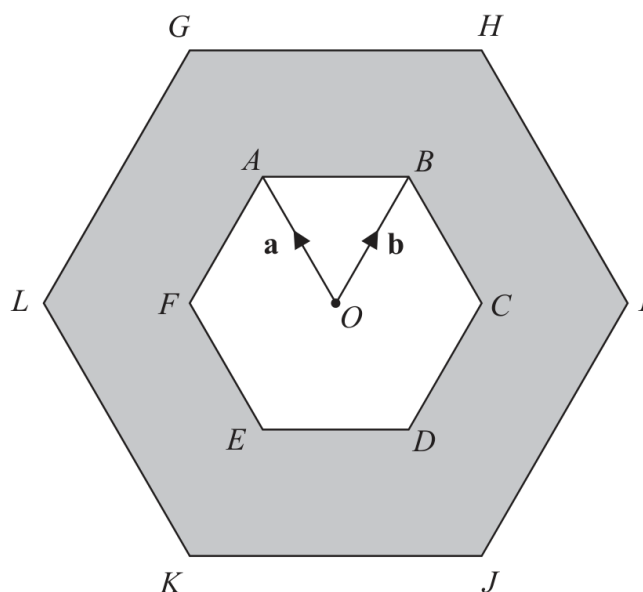


Diagram **NOT**
accurately drawn

$GHIJKL$ is an enlargement of $ABCDEF$, with centre O and scale factor 2

$$\vec{OA} = \mathbf{a} \quad \vec{OB} = \mathbf{b}$$

(a) Write the following vectors, in terms of $\mathbf{a}$ and $\mathbf{b}$.
Simplify your answers.

(i) $\vec{AB}$

(1)

(ii) $\vec{KI}$

(2)

(iii) $\vec{LD}$

(2)

The triangle OAB has an area of 5 cm^2

(b) Calculate the area of the shaded region.

(3)

cm^2

Worked solution

November 2021 | Question 14 | 2 marks

Family: Calculate with vectors | Variant: Find a resultant along several moves

METHOD / WORKING

1. (a)(i)
2. $\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA}$
3. $= \mathbf{b} - \mathbf{a}$
4. For a regular hexagon,
5. $\overrightarrow{OC} = \mathbf{b} - \mathbf{a}, \quad$
6. $\overrightarrow{OD} = -\mathbf{a}, \quad$
7. $\overrightarrow{OE} = -\mathbf{b}, \quad$
8. $\overrightarrow{OF} = \mathbf{a} - \mathbf{b}$
9. The outer hexagon has scale factor 2, so
10. $\overrightarrow{OI} = 2(\mathbf{b} - \mathbf{a}), \quad$
11. $\overrightarrow{OK} = -2\mathbf{b}, \quad$
12. $\overrightarrow{OL} = 2(\mathbf{a} - \mathbf{b})$

FINAL ANSWER

(ii) $4\mathbf{b} - 2\mathbf{a}$

14 $ABCDEF$ and $GHIJKL$ are regular hexagons each with centre O .

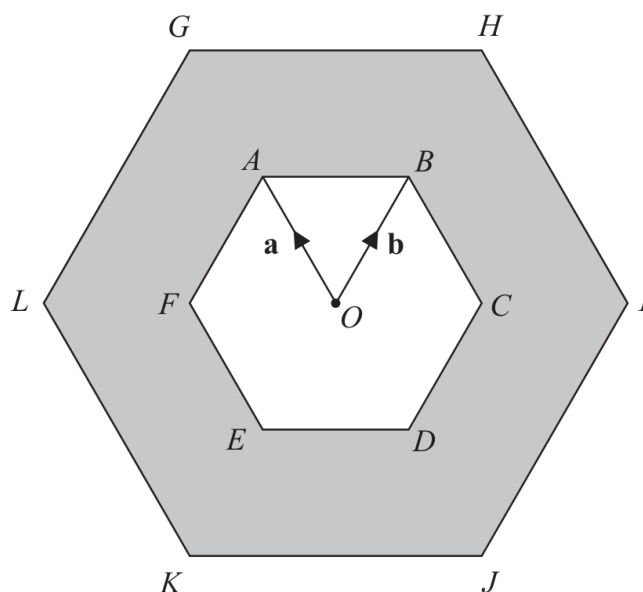


Diagram **NOT**
accurately drawn

$GHIJKL$ is an enlargement of $ABCDEF$, with centre O and scale factor 2

$$\vec{OA} = \mathbf{a} \quad \vec{OB} = \mathbf{b}$$

(a) Write the following vectors, in terms of $\mathbf{a}$ and $\mathbf{b}$.
Simplify your answers.

(i) $\vec{AB}$

(1)

(ii) $\vec{KI}$

(2)

(iii) $\vec{LD}$

(2)

The triangle OAB has an area of 5 cm^2

(b) Calculate the area of the shaded region.

(3)

cm^2

Worked solution

November 2021 | Question 14 | 2 marks

Family: Use vectors with ratios and intersections • Variant: Use vectors in a parallelogram, trapezium or polygon path

Method / working

1. (ii)

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

2. $\overrightarrow{KI}$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

3. $=\overrightarrow{OI}-\overrightarrow{OK}$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

4. $=2(\mathbf{b}-\mathbf{a})-(-2\mathbf{b})$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

5. $=4\mathbf{b}-2\mathbf{a}$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

Final answer: (iii) $2\mathbf{b}-3\mathbf{a}$

19 OAB is a triangle.

$$\vec{OA} = \mathbf{a} \quad \vec{OB} = \mathbf{b}$$

The point C lies on OA such that $OC : CA = 1 : 2$

The point D lies on OB such that $OD : DB = 1 : 2$

Using a vector method, prove that $ABDC$ is a trapezium.

(Total for Question 19 is 3 marks)

Answer reference | May-Nov 2020 | Question 19

Family: Use vectors with ratios and intersections • Variant: Use vectors in a parallelogram, trapezium or polygon path

19	$\overrightarrow{AB} = -\mathbf{a} + \mathbf{b}$ or $\overrightarrow{BA} = \mathbf{a} - \mathbf{b}$			M1	Correct diagram (condone missing vector labels or arrows – with C on line segment OA and D on line segment OB) OR for finding $\overrightarrow{AB}$ or $\overrightarrow{BA}$ - may be seen as part of later working
	$\overrightarrow{CD} = \frac{1}{3}(-\mathbf{a} + \mathbf{b})$ or $\overrightarrow{DC} = \frac{1}{3}(\mathbf{a} - \mathbf{b})$ oe			M1	Method to find $\overrightarrow{CD}$ or $\overrightarrow{DC}$
		Correct vectors and conclusion including <u>parallel</u> and <u>trapezium</u>	3	A1	eg $\overrightarrow{AB}$ (AB) and $\overrightarrow{CD}$ (CD) are parallel therefore $ABDC$ is a trapezium
Total 3 marks					

Worked solution

May-Nov 2020 | Question 19 | 3 marks

Family: Use vectors with ratios and intersections • Variant: Use vectors in a parallelogram, trapezium or polygon path

Official final mark-scheme pages are embedded immediately after this question.

24 $OAED$ is a quadrilateral.

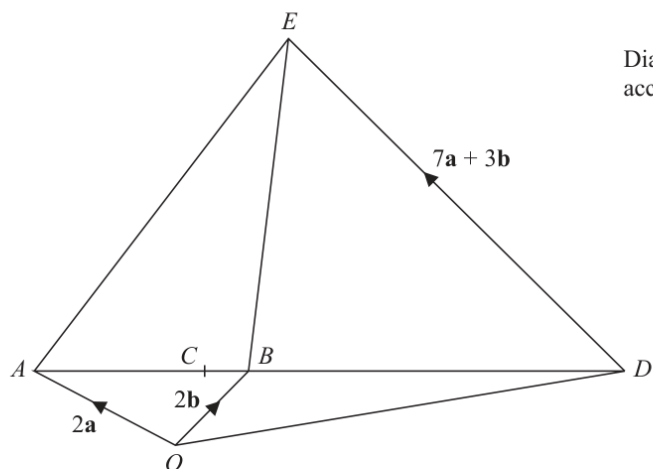


Diagram **NOT**
accurately drawn

$$\vec{OA} = 2\mathbf{a} \quad \vec{OB} = 2\mathbf{b} \quad \vec{DE} = 7\mathbf{a} + 3\mathbf{b}$$

$$AB:BD = 1:2$$

The point C on AB is such that OCE is a straight line.

Use a vector method to find the ratio of $OC:CE$



(Total for Question 24 is 5 marks)

Answer reference | January 2023 | Question 24

Family: Use vectors with ratios and intersections • Variant: Find a section or midpoint vector

24	e.g. $(\overrightarrow{AB}) = 2\mathbf{b} - 2\mathbf{a}$ oe or $(\overrightarrow{BA}) = 2\mathbf{a} - 2\mathbf{b}$ oe or $(\overrightarrow{BD}) = 2(2\mathbf{b} - 2\mathbf{a}) (= 4\mathbf{b} - 4\mathbf{a})$ oe or $(\overrightarrow{AD}) = 3(2\mathbf{b} - 2\mathbf{a}) (= 6\mathbf{b} - 6\mathbf{a})$ oe		5	M1
	e.g. $(\overrightarrow{OE}) = 2\mathbf{b} + 2(2\mathbf{b} - 2\mathbf{a}) + 7\mathbf{a} + 3\mathbf{b} (= 3\mathbf{a} + 9\mathbf{b})$ oe or $(\overrightarrow{OC}) = 2\mathbf{a} + \lambda(2\mathbf{b} - 2\mathbf{a}) = (2 - 2\lambda)\mathbf{a} + 2\lambda\mathbf{b}$ oe or $2\mathbf{b} + \lambda(2\mathbf{a} - 2\mathbf{b})$ or $(\overrightarrow{CE}) = (2\mathbf{b} - 2\mathbf{a}) - \lambda(2\mathbf{b} - 2\mathbf{a}) + 2(2\mathbf{b} - 2\mathbf{a}) + 7\mathbf{a} + 3\mathbf{b} = (1 + 2\lambda)\mathbf{a} + (9 - 2\lambda)\mathbf{b}$			M2 for 2 correct paths seen M1 for 1 correct path seen Any correct path for OC passing through A or B involving a variable
	e.g. $\frac{2 - 2\lambda}{2\lambda} = \frac{1 + 2\lambda}{9 - 2\lambda}$ oe or $\frac{2 - 2\lambda}{2\lambda} = \frac{3}{9}$ oe or $\frac{(1 + 2\lambda)}{(9 - 2\lambda)} = \frac{1}{3}$ oe or $\lambda = \frac{3}{4}$ or $(2 - 2\lambda)\mathbf{a} + 2\lambda\mathbf{b} = \mu(3\mathbf{a} + 9\mathbf{b})$ or $\lambda = \frac{3}{4}$ or $\mu = \frac{1}{6}$ or $2\mathbf{b} + \lambda(2\mathbf{a} - 2\mathbf{b}) = \mu(3\mathbf{a} + 9\mathbf{b})$ or $\lambda = \frac{1}{4}$ or $\mu = \frac{5}{6}$			M1 for comparing coefficients of $\mathbf{a}$ and $\mathbf{b}$ for $(OC \text{ and } CE)$ or $(OC \text{ and } OE)$ or $(CE \text{ and } OE)$ OC is a multiple of OE Two different paths for OC
		1 : 5		A1 dep M2 oe e.g 2 : 10
	Working required			Total 5 marks

Answer reference | January 2023 | Question 24

Family: Use vectors with ratios and intersections • Variant: Find a section or midpoint vector

24 ALT	e.g. $(\overrightarrow{AB}) = 2\mathbf{b} - 2\mathbf{a}$ oe or $(\overrightarrow{BA}) = 2\mathbf{a} - 2\mathbf{b}$ oe or $(\overrightarrow{BD}) = 2(2\mathbf{b} - 2\mathbf{a}) (= 4\mathbf{b} - 4\mathbf{a})$ oe or $(\overrightarrow{AD}) = 3(2\mathbf{b} - 2\mathbf{a}) (= 6\mathbf{b} - 6\mathbf{a})$ oe		5	M1
	e.g. $(\overrightarrow{OE}) = 2\mathbf{b} + 2(2\mathbf{b} - 2\mathbf{a}) + 7\mathbf{a} + 3\mathbf{b} (= 3\mathbf{a} + 9\mathbf{b})$ oe			M1
	e.g. $(\overrightarrow{AE}) = 2\mathbf{b} - 2\mathbf{a} + 2(2\mathbf{b} - 2\mathbf{a}) + 7\mathbf{a} + 3\mathbf{b} (= \mathbf{a} + 9\mathbf{b})$ oe			M1
	$[\overrightarrow{AE} = \lambda \overrightarrow{AD} + \mu \overrightarrow{OE}]$ $\mathbf{a} + 9\mathbf{b} = \lambda(6\mathbf{b} - 6\mathbf{a}) + \mu(3\mathbf{a} + 9\mathbf{b})$ oe or $1 = -6\lambda + 3\mu$ oe and $9 = 6\lambda + 9\mu$ oe or or $\lambda = \frac{1}{4}$ or $\mu = \frac{5}{6}$			M1
		1 : 5		A1 dep on M2 oe e.g 2 : 10
	Working required			Total 5 marks

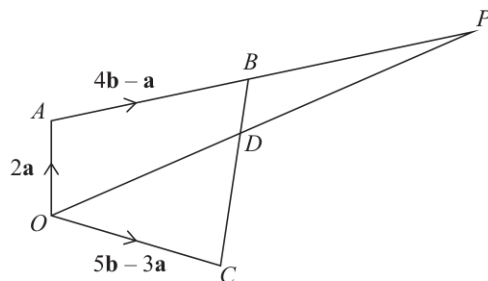
Worked solution

January 2023 | Question 24 | 5 marks

Family: Use vectors with ratios and intersections • Variant: Find a section or midpoint vector

Official final mark-scheme pages are embedded immediately after this question.

27

Diagram **NOT**
accurately drawn $OABC$ is a quadrilateral. ABP and ODP are straight lines.

$$\vec{OA} = 2\mathbf{a} \quad \vec{AB} = 4\mathbf{b} - \mathbf{a} \quad \vec{OC} = 5\mathbf{b} - 3\mathbf{a}$$

- (a) Find an expression in terms of $\mathbf{a}$ and $\mathbf{b}$ for the vector $\vec{BC}$
Simplify your answer.

(2)

The point D lies on BC such that $BD:DC = 1:3$ Given that $\vec{OP} = n\vec{OD}$

- (b) use a vector method to find the value of n

24



$$n = \dots\dots\dots (4)$$

(Total for Question 27 is 6 marks)

Worked solution

November 2025 | Question 27 | 6 marks

Family: Use vectors with ratios and intersections • Variant: Find an intersection vector from two paths

Method

$$\vec{OA} = 2\mathbf{a}, \quad \vec{AB} = 4\mathbf{b} - \mathbf{a}, \quad \vec{OC} = 5\mathbf{b} - 3\mathbf{a}$$

First find $\vec{OB}$:

$$\vec{OB} = \vec{OA} + \vec{AB}$$

$$\vec{OB} = 2\mathbf{a} + 4\mathbf{b} - \mathbf{a}$$

$$\vec{OB} = \mathbf{a} + 4\mathbf{b}$$

For part (a),

$$\vec{BC} = \vec{OC} - \vec{OB}$$

$$\vec{BC} = (5\mathbf{b} - 3\mathbf{a}) - (\mathbf{a} + 4\mathbf{b})$$

$$\vec{BC} = \mathbf{b} - 4\mathbf{a}$$

For part (b), $BD:DC = 1:3$, so

$$\vec{OD} = \vec{OB} + \frac{1}{4}\vec{BC}$$

$$\vec{OD} = \mathbf{a} + 4\mathbf{b} + \frac{1}{4}(\mathbf{b} - 4\mathbf{a})$$

$$\vec{OD} = \frac{3}{4}\mathbf{a} + \frac{17}{4}\mathbf{b}$$

Since A, B, P are collinear, let

$$\vec{OP} = \vec{OA} + s\vec{AB}$$

$$\vec{OP} = 2\mathbf{a} + s(4\mathbf{b} - \mathbf{a})$$

$$\vec{OP} = (2-s)\mathbf{a} + 4s\mathbf{b}$$

But O, D, P are collinear, and $\vec{OD}$ has only a $\mathbf{b}$ component.

So

$$2-s=0$$

$$s=2$$

Then

$$\vec{OP} = 8\mathbf{b}$$

Given

$$\vec{OP} = n\vec{OD}$$

$$8\mathbf{b} = n\left(\frac{3}{4}\mathbf{a} + \frac{17}{4}\mathbf{b}\right)$$

$$n = \frac{32}{17}$$

$$\text{Answer: } \vec{BC} = \mathbf{b} - 4\mathbf{a}, \quad n = \frac{32}{17}$$